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Objective

Our plan is to open a cafe in a new location. We have 4 possible locations (A~D) that are being considered, and we want to find the most optimal choice for our business. We will use the Monte Carlo simulation to assess the risks and rewards associated with each location to figure out which area will be best for opening a cafe chain.

The provided data

Given 4 locations, each with the following information:

- estimated initial investment
- estimated monthly operating costs
- estimated monthly foot traffic (number of potential customers)
- estimated average spend per customer

The cafe chain's required rate of return (discount rate) is 10%.

The company's target payback period is 3 years.

Probability Distributions

- Initial Investment: Triangular distribution with min, max, and most likely values provided for each location.
- Monthly operating costs: Normal distribution with mean provided for each location.
- Monthly foot traffic: Poisson distribution with mean provided for each location.
- Average spend per customer: Lognormal distribution with mean and standard deviation provided for each location.

Tasks.

- T1. Create Monte Carlo Simulation model for each potential location. Calculate the monthly revenue and net cash flow for each location.
- T2. Calculate the Net Present Value for each location using the cafe chain's required rate of return.
- T3. Determine the payback period for each location based on the cumulative net cash flow.
- T4. Calculate the Return on Investment for each location after the target payback period of 3 years.
- T5. Analyze the results of your simulations, considering NPV. Identify the optimal location for expansion based on your analysis.
- T6. Discuss the risk and uncertainties associated with the optimal location and any potential mitigation strategies
- T7. Summary of findings and recommendations to the management team.

Monte Carlo Simulation model for each location.

```
set.seed(123)
num_sim = 1000

# load the dataset
data <- read_excel("Case4_distParameters.xlsx")

# we are given target payback period = 3 years
# since we have data based on monthly costs/revenue/etc we convert it to months
num_months = 36

# we are given that the rate of return is 10%. We assume this is annual.
discount_rate = 0.10

# the number of cafe locations we have data on
num_loc <- nrow(data)

# we need to convert annual discount rate to monthly
# Monthly Rate =  $(1 + \text{Annual Rate})^{1/12} - 1$  (math formula taken from the web)
monthly_discount_rate = (1 + discount_rate)^(1/12) - 1

# net-present profit model per month
netPresentProfit <- function(dr, profit, target_month) {
  netProfit <- 0
  for (cur_month in 1:target_month){
    netProfit <- netProfit + profit[cur_month] / (1+dr)^cur_month
  }
  return(netProfit)
}

# create empty matrix of simulated NPV for all 4 cafe locations
# row = 1000 (number of simulations)
# col = 4 (cafe locations)
simulated_npv_matrix <- matrix(nrow = num_sim, ncol = num_loc)

# create empty 3d array of simulated cumulative net cash flow
# row = 36 (number of months in 3 year payback period)
# col = 4 (number of cafe locations)
# matrix = 1000 (number of simulations)
simulated_cum_ncf_array <- array(dim=c(num_months, num_loc, num_sim))

for (i in 1:num_sim){
  # initial investment~triangular probability distribution for each location A~D
  # from every row in dataset which represents a different cafe location get parameters
  # parameter = min, max, and most likely initial investment cost
  # rtriangle(1, min, max, most likely)
  initial_investments <- apply(data, 1, function(location) {
    min_val = as.numeric(location[2])
    max_val = as.numeric(location[4])
    most_likley = as.numeric(location[3])
    rtriangle(1, min_val, max_val, most_likley)
  })
}
```

```

# monthly operating costs ~ normal probability distribution for each location A~D
# from every row in dataset which represents a different cafe location get parameters
# parameters = mean and sd of operating cost
# rnorm(36, mean, sd)
monthly_opcosts <- apply(data, 1, function(location) {
  mean_val = as.numeric(location[5])
  sd_val = as.numeric(location[6])
  rnorm(num_months, mean_val, sd_val)
})

# monthly foot traffic ~ poisson probability distribution for each location A~D
# from every row in dataset which represents a different cafe location get parameters
# parameters = mean of monthly foot traffic
# rpois(36, mean)
monthly_foottraffic <- apply(data, 1, function(location) {
  mean_val = as.numeric(location[7])
  rpois(num_months, mean_val)
})

# function that convert the mean and sd into log normal parameters
# we get inaccurate values if we do not convert the parameters
# used the web to get function to convert arithmetic mean/sd to log-normal mean/sd
get_log <- function(mean, sd) {
  variance <- sd^2
  log_sd <- sqrt(log(1 + (variance / (mean^2))))
  log_mean <- log(mean) - (0.5 * (log_sd^2))
  return(list(log_mean = log_mean, log_sd=log_sd))
}

# Average spend per customer ~ lognormal probability distribution for each location A~D
# from every row in dataset which represents a different cafe location get parameters
# parameters = lognormal mean and sd of average spend per customer
# rlnorm(36, log_mean, log_sd)
avg_spendpc <- apply(data, 1, function(location) {
  mean_val <- as.numeric(location[8])
  sd_val <- as.numeric(location[9])
  # use get_log to get log parameters
  log_val <- get_log(mean_val, sd_val)
  rlnorm(num_months, log_val$log_mean, log_val$log_sd)
})

# calculate monthly revenue for each cafe location A~D
# monthly revenue = monthly foot traffic * average spend per customer
monthly_revenues <- monthly_foottraffic * avg_spendpc

# calculate monthly net cash flow for each cafe location A~D
# monthly net cash flow = monthly rev - monthly operation cost
monthly_ncf <- monthly_revenues - monthly_opcosts

# get NPV profits for each location (A~D) using netPresentProfit model
npvProfits <- apply(monthly_ncf, 2, function(location_ncf){
  netPresentProfit(monthly_discount_rate, location_ncf, num_months)
})

```

```

# store all simulated npv for each location in our empty matrix
# we take into account our initial investment expense
# each row (i) represents a simulation
simulated_npv_matrix[i, ] <- npvProfits - initial_investments

# get the cumulative profit for each month of each location (A~D)
# row = number of months
# col = number of locations
cumulative_profits <- apply(monthly_ncf, 2, cumsum)

# make vector initial_investments into matrix
init_inv_matrix <- matrix(initial_investments, nrow = num_months, ncol = num_loc, byrow = TRUE)

# for every simulation, we want cumulative net cash flow for all 4 locations
# store it inside 3d array
simulated_cum_ncf_array[ , , i] <- cumulative_profits - init_inv_matrix
}

```

We take the mean of our simulated NPV for each location to find the expected NPV.

```

# expected npv of location A after simulation and its sd
mean(simulated_npv_matrix[,1])

```

```
## [1] 478743.7
```

```
sd(simulated_npv_matrix[,1])
```

```
## [1] 37044.14
```

The cafe in location A has expected NPV of \$478,743.7.

```

# expected npv of location B after simulation and its sd
mean(simulated_npv_matrix[,2])

```

```
## [1] 576708.2
```

```
sd(simulated_npv_matrix[,2])
```

```
## [1] 37598.93
```

The cafe in location B has expected NPV of \$576,708.2.

```

# expected npv of location C after simulation and its sd
mean(simulated_npv_matrix[,3])

```

```
## [1] 912212.9
```

```
sd(simulated_npv_matrix[,3])
```

```
## [1] 53832.08
```

The cafe in location C has expected NPV of \$912,212.9.

```
# expected npv of location D after simulation and its sd  
mean(simulated_npv_matrix[,4])
```

```
## [1] 527304.3
```

```
sd(simulated_npv_matrix[,4])
```

```
## [1] 32333.43
```

The cafe in location D has expected NPV of \$527,304.3.

Using cumulative net cash flow, we can find the payback period for each location.

```
# create function to get the month, the cafe recovered its initial investment cost.  
targetpayback_func <- function(matrix) {  
  # dimension of matrix  
  simulation_len = ncol(matrix)  
  payback_period = nrow(matrix)  
  # empty vector of simulation length to store results  
  simulated_payback <- numeric(simulation_len)  
  # find the month cumulative ncf was >= 0 during cafe's 36 month (3 year) target payback period.  
  for (cur_sim in 1:simulation_len){  
    month_broke_even <- 0  
    for (cur_mon in 1:payback_period) {  
      if (matrix[cur_mon,cur_sim] >=0) {  
        # we found the month the cafe made it out of negative and we exit inner loop  
        month_broke_even <- cur_mon  
        break  
      }  
    }  
    simulated_payback[cur_sim] <- month_broke_even  
  }  
  return(simulated_payback)  
}  
  
# for each location, get all simulated payback months  
simulated_paybacks <- apply(simulated_cum_ncf_array, 2, function(matrix){  
  targetpayback_func(matrix)  
})
```

We take the mean of all the simulated_payback months (the month the cafe broke past the negative) for each cafe location across all simulations.

```
colMeans(simulated_paybacks)
```

```
## [1] 8.911 6.620 6.374 6.329
```

We get that cafe in location A has expected payback period of 8.911 months. Cafe in location B has expected payback period of 6.620 months. Cafe in location C has expected payback period of 6.374 months. Cafe in location D has expected payback period of 6.329 months.

To calculate the Return on Investment (ROI) for each location after the target payback period of 3 years we need to use formula $ROI = (\text{net return} / \text{cost of investment}) * 100$.

```
# 4x1000 matrix that represents cumulative total ncf which is at the 36th month
simulated_net_return <- simulated_cum_ncf_array[36, , ]

# apply ROI formula
simulated_roi <- (simulated_net_return / initial_investments) * 100

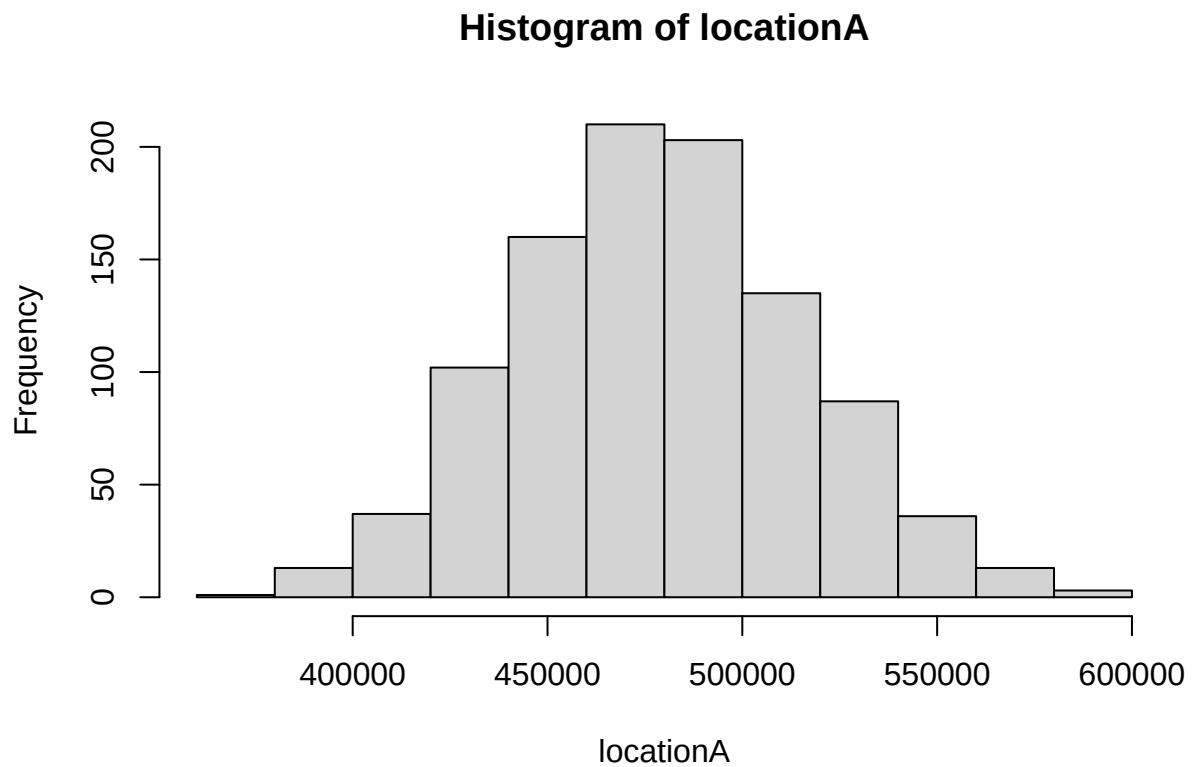
# get mean of all simulated ROI for each cafe location (A~D in that order)
rowMeans(simulated_roi)
```

```
## [1] 362.7164 444.7199 520.6896 529.4645
```

The cafe in location A has expected ROI of 362.7164%. The cafe in location B has expected ROI of 444.7199 %. The cafe in location C has expected ROI of 520.6896%. The cafe in location D has expected ROI of 529.4645%.

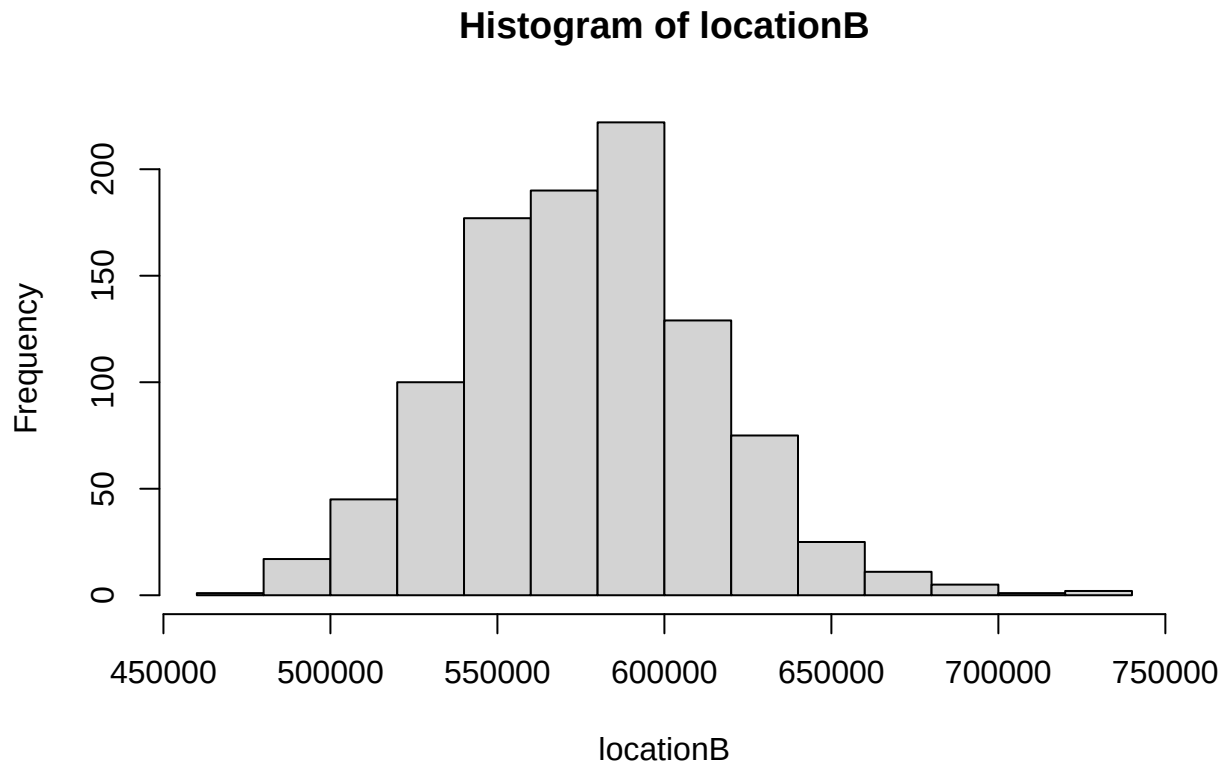
Analysis of the results of our simulations, considering NPV to identify the optimal location for expansion based on analysis.

```
# simulated npv histogram of cafe in location A
locationA <- simulated_npv_matrix[,1]
hist(locationA)
```



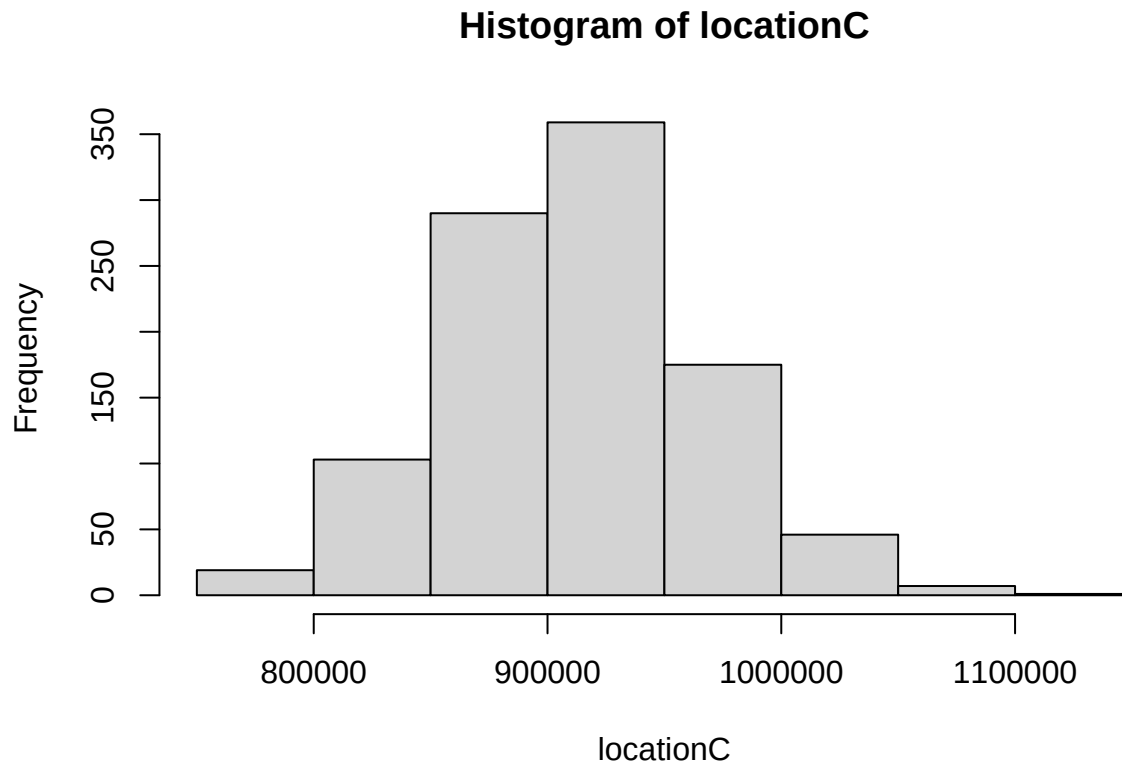
The histogram of location A indicates the majority of expected NPV is distributed between \$400,000 and \$550,000.

```
# simulated npv histogram of cafe in location B  
locationB <- simulated_npv_matrix[,2]  
hist(locationB)
```



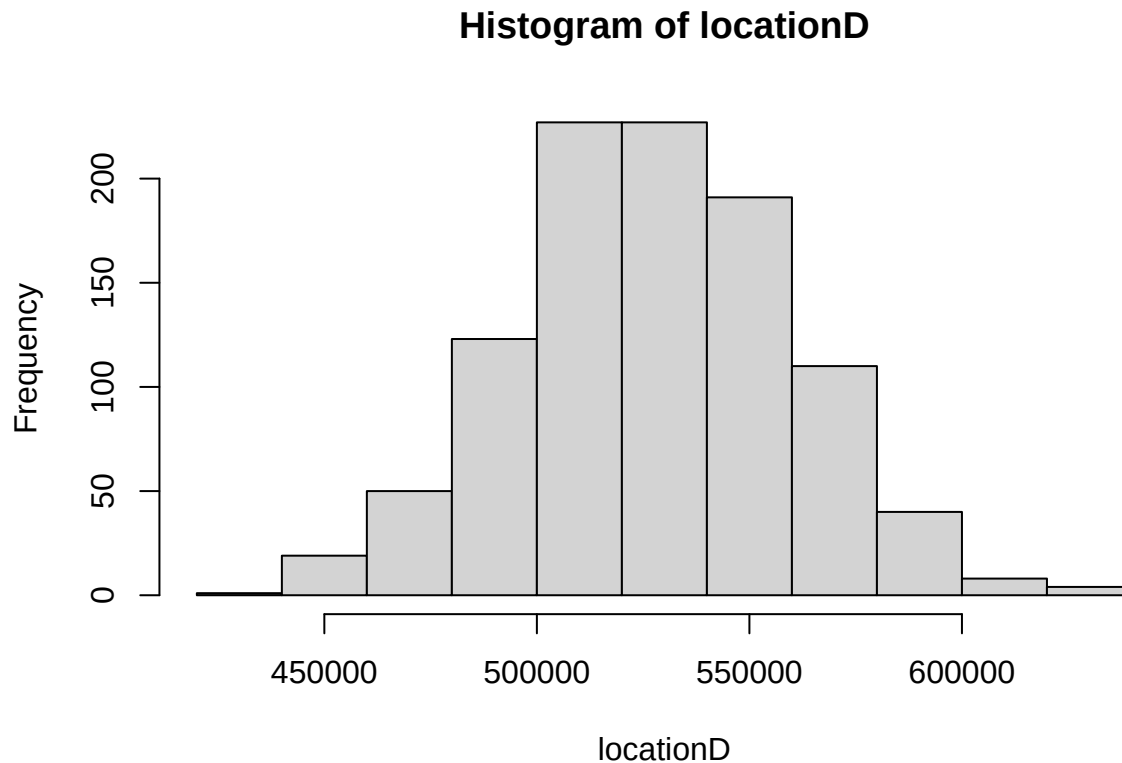
The histogram of location B indicates the majority of expected NPV is distributed between \$500,000 and \$650,000. Location B seems to have a bigger range of possible expected NPV compared to the other locations which indicates high variance.

```
# simulated npv histogram of cafe in location C
locationC <- simulated_npv_matrix[,3]
hist(locationC)
```



The histogram of location C indicates the majority of expected NPV is distributed between \$800,000 and \$1,000,000. Location C seems to be the most stable among the 4 locations as it has the least fluctuations in terms of possible expected NPV values.

```
# simulated npv histogram of cafe in location D
locationD <- simulated_npv_matrix[,4]
hist(locationD)
```



The histogram of location D indicates the majority of expected NPV is distributed between \$450,000 and \$600,000.

Taking into account the expected NPV, histogram of simulated NPV, expected ROI, and the expected target payback period, location C seems like the most optimal choice for expansion. Cafe in location C by far has the highest expected NPV value at \$912,212.9 which is \$335,504.7 higher than the next highest expected NPV value from cafe in location B with expected NPV of \$576,708.2. Looking at the histogram, cafe in location C has the majority of its expected NPV between \$800,000 and \$1,000,000 which is the highest expected NPV distribution among the cafe locations. Both location C and D has the lowest expected payback period at 6.374 months and 6.329 months respectively. However, given that location C having the highest, most likely initial investment cost at \$210,000 is nearly double that of location D with its most likely initial investment cost at \$120,000 this looks very favorable for location C. The expected ROI of location C being 520.6896% which is the second highest among the 4 locations after 529.4645% from location D indicates that even with high costs in the initial investment, the location is good enough to make up for it in revenue over time. Cafe expansion in location C would be the most optimal choice given the data.

Any possible risks?

First, with location C we have to take into account that we have not only have the highest possible initial investment costs, but also the highest possible operating costs as well. The only reason why location C seems optimal despite having the highest costs (operating + initial investment) is due to the assumption that it will make up for the expenses in significantly higher revenue compared to the other locations.

Another point of worry is that we are only given the mean of foot traffic for each location so it

is a possibility that there are periods during the year where foot traffic spikes high and other times where the actual foot traffic is much less than the expected mean.

A possible solution to these risk could include lowering the operating costs. Instead of having a roster of full time staff, a mix of full time and part time workers could make the cafe more flexible to handle periods where revenue might not be as high due to low foot traffic or lower than expected average spend per customer. A percentage rent agreement is also an option so that you're not paying as much rent during periods where the cafe isn't performing as well as expected.

Summary & Recommendations

Based on the data and running it through our model we have concluded that location C is the most optimal area to expand our cafe businesses. Cafe in location C has the highest expected Net Present Value (NPV) at \$912,212.9. This is \$335,504.7 higher than the next highest expected NPV value from cafe in location B with expected NPV of \$576,708.2. Even if we take -3 standard deviations of location C's NPV, location C still has an expected NPV of \$750,716.66 which is higher than any of the other location's expected NPV.

The expected payback period for location C is 6.374 months which is only second lowest if the four locations after 6.329 months for location D. However, if we take into account that the initial investment cost and monthly operating costs for location C is nearly double that of location D, this shows that the potential revenue of Location C is big enough to overcome its high costs in roughly the same time it takes a much cheaper location to break even.

The expected ROI for location C is 520.6896%. Compare this to the highest ROI of 529.4645% from location D, there is only a very small difference. But if we take the most likely initial investment costs of location C which is \$210,000 and compare that to the most likely initial investment costs of location D which is \$120,000. The profit for location C is \$1,093,448.16 while the profit for location D is \$635,357.4. After the target payback period of 3 years, by choosing location C, we are getting \$458,090.76 more dollars which is big difference in profit.

Overall, based on what the model showed us, we believe that location C would be the best location for the cafe business to expand to.